Reinforcement Learning
LAB 1

k-armed Bandits

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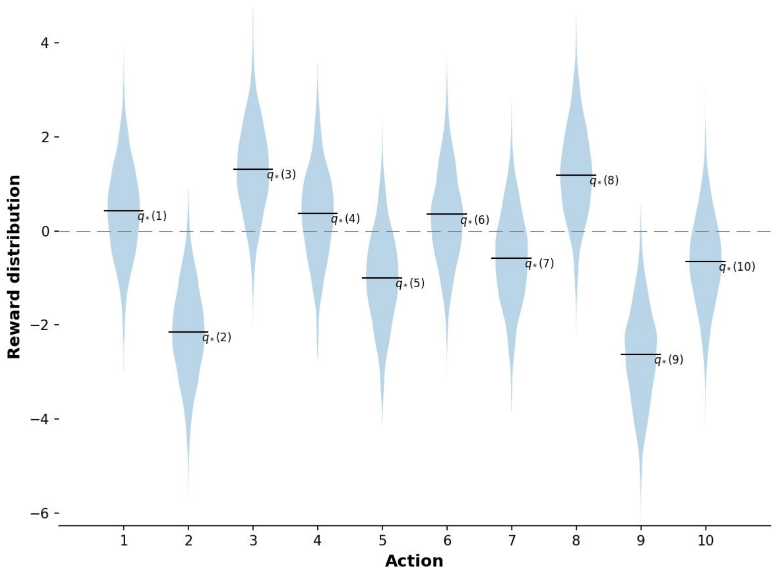
Quick recap

1. K bandits (slot machines)
2. each bandit has a stochastic reward function
3. we can try them as many times as we want
4. the aim is to find which one is the best, trying to avoid spending billions



Components

Policy

Given an estimate of the *Q function*, which action should I choose?

Update

Given an estimate of the *Q function* and a new *reward R*, how should I update *Q*?

```
Policy(policy_update, actions, inital_value)
    act() # which action to pick
    step(action, reward) # update belief of Q
    reset() # reset estimates
```

```
EpsilonGreedyPolicy        UCBPolicy        GradientPolicy
```

EpsGreedyPolicy

$$A_t = \begin{cases} argmax_{a \in A}\ q(a) & \text{with } P = 1 - \epsilon \\ \text{uniform random with } P = \epsilon \end{cases}$$

Update:

- Sample average: $$Q_{t+1}(a) = Q_t(a) + \frac{1}{n}(R_t - Q_t(a))$$
- Running average: $$Q_{t+1}(a) = Q_t(a) + \alpha(R_t - Q_t(a))$$

UCBPolicy

$$A_t = argmax_{a \in A} \left[Q_t(a) + c\sqrt{\frac{\ln t}{N_t(a)}} \right]$$

Update:

- Sample average: $$Q_{t+1}(a) = Q_t(a) + \frac{1}{n}(R_t - Q_t(a))$$
- Running average: $$Q_{t+1}(a) = Q_t(a) + \alpha(R_t - Q_t(a))$$

GradientPolicy

$$P\{A_t = a\} = \frac{e^{H_t(a)}}{\sum_{a'} e^{H_t(a')}}$$

Update:

- Gradient update: $$\begin{cases} H(A_t) = H(A_t) + \alpha(R - \bar{R})(1 - \pi_t(A_t)) \\ H(a) = H(a) - \alpha(R - \bar{R})\pi_t(a), \ \forall a \neq A_t \end{cases}$$

Why is this formula like this?

$$\begin{cases} H(A_t) = H(A_t) \underline{+} \alpha(R - \bar{R})\underline{(1 - \pi_t(A_t))} \\ H(a) = H(a) \underline{-} \alpha(R - \bar{R})\underline{\pi_t(a)}, \ \forall a \neq A_t \end{cases}$$

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$$\begin{cases} H(A_t) = H(A_t) + \alpha(R - \bar{R})(1 - \pi_t(A_t)) \\ H(a) = \quad H(a) + \quad \alpha(R - \bar{R})(0 - \pi_t(a)), \ \forall a \neq A_t \end{cases}$$

Why is this formula like this?

We can see this as a weighted regression task, where the weight is $(R - \bar{R})$ which, if negative, has a repulsive effect

$$\begin{cases} H(A_t) = H(A_t) + \alpha(R - \bar{R})(1 - \pi_t(A_t)) \\ H(a) = H(a) - \alpha(R - \bar{R})\pi_t(a), \ \forall a \neq A_t \end{cases}$$

$$\begin{cases} H(A_t) = H(A_t) + \alpha(R - \bar{R})(1 - \pi_t(A_t)) \\ H(a) = \ H(a) + \ \alpha\underbrace{(R - \bar{R})}_{w}\underbrace{(0 - \pi_t(a))}, \ \forall a \neq A_t \end{cases}$$

$$H = H - \alpha \cdot w \cdot \nabla(y - \pi(a))^2$$

$$\frac{\nabla(y - \hat{y})^2}{2} \text{ with } y \in \{0, 1\}, \hat{y} = \pi(a)$$

Example

Say we have a 3 armed bandits, and we start with a uniform prior over them.
Since GBA uses softmax as transformation of the logits, any uniform initialization will induce a uniform distribution, so say we start with:

$$H(a) = 1, \ \forall a \in A \qquad \bar{R} = 3$$

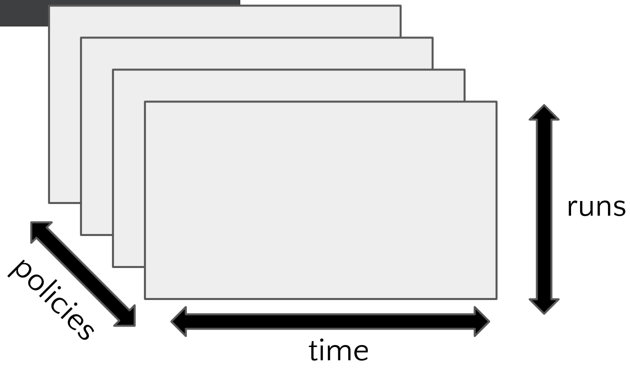
Say we pull the second bandit and we get a reward of 4:

$$\begin{cases} H(A_t) = H(A_t) + \alpha(R - \bar{R})(1 - \pi_t(A_t)) \\ H(a) = H(a) - \alpha(R - \bar{R})\pi_t(a), \ \forall a \neq A_t \end{cases} \Rightarrow \begin{cases} H(A_t) = H(A_t) + \alpha(4-3)(1-0.\bar{3}) \\ H(a) = \ H(a) + \ \alpha(4-3)(0-0.\bar{3}), \ \forall a \neq A_t \end{cases}$$

$$\begin{cases} H(A_t) = H(A_t) + \alpha \cdot 0.\bar{6} \\ H(a) = \ H(a) \ - \alpha \cdot 0.\bar{3}, \ \forall a \neq A_t \end{cases}$$

Simulation

```
def simulate(runs, time, policies: List[Policy], environment: Environment):
    rewards = np.zeros((len(policies), runs, time))
    best_action_counts = np.zeros(rewards.shape)
    for i, policy in enumerate(policies):
        for r in trange(runs):
            policy.reset()
            environment.reset()
```



Simulation

```python
def simulate(runs, time, policies: List[Policy], environment: Environment):
    rewards = np.zeros((len(policies), runs, time))
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    for i, policy in enumerate(policies):
        for r in trange(runs):
            policy.reset()
            environment.reset()
            for t in range(time):
                action = policy.act()
                reward = environment.reward(action)
                policy.step(action, reward)
                rewards[i, r, t] = reward
```

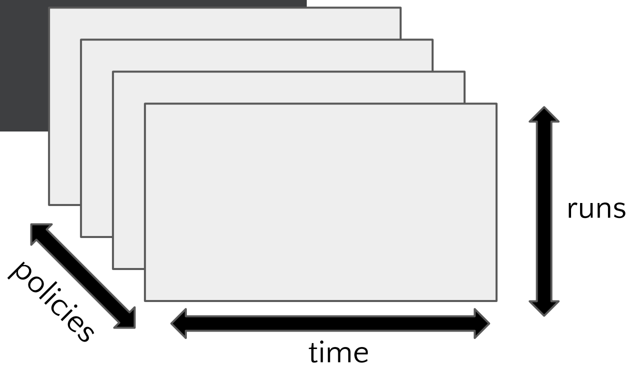

Simulation

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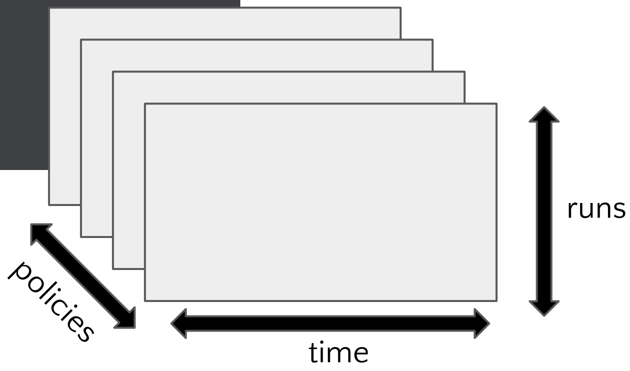
                reward = environment.reward(action)
                policy.step(action, reward)
                rewards[i, r, t] = reward
                if action == environment.best_action:
                    best_action_counts[i, r, t] = 1
```



Simulation

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                reward = environment.reward(action)
                policy.step(action, reward)
                rewards[i, r, t] = reward
                if action == environment.best_action:
                    best_action_counts[i, r, t] = 1
    mean_best_action_counts = best_action_counts.mean(axis=1)
    mean_rewards = rewards.mean(axis=1)
    return mean_best_action_counts, mean_rewards
```

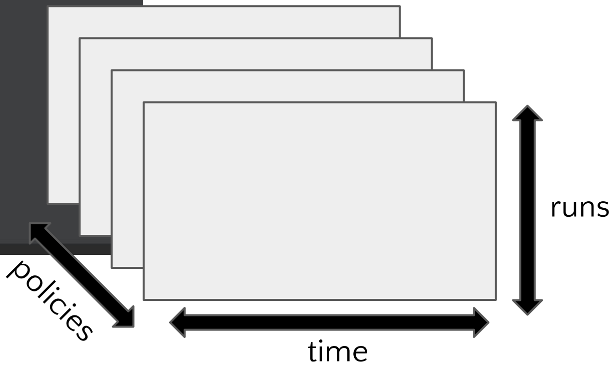

You can play around with policies

Consider a setting where each bandit either wins or loses, so the reward is either 1 or -1

Say you have 2 bandits, and the first one you tried it 10 times, and won 6 times, and the second one you tried it 100 times, and won 70 times.

What would you do?

You can play around with policies

Consider a setting where each bandit either wins or loses, so the reward is either 1 or -1

Say you have 2 bandits, and the first one you tried it 10 times, and won 6 times, and the second one you tried it 100 times, and won 70 times.

What would you do?

Even though the second one has a higher percentage rate of wins, the second one has still some “chance” that the bad results was bad luck right?

Bayes theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

If the likelihood and the prior are conjugate, we can calculate the posterior in closed form

Thompson Policy

Assume we know the reward distribution family:

$$r(a) \sim N(\mu, \sigma^2)$$

And for the sake of simplicity, assume we know

$$\sigma = 1$$

Then, we can use a Bayesian process to regress such distribution*

*for closed form calculations, refer to https://www.cs.ubc.ca/~murphyk/Papers/bayesGauss.pdf

Thompson Policy

Once we know the posterior distribution for each bandit, we can sample from it

$$q(a_i) \sim N(\mu_i, \sigma_i^2)$$

And then we can take the greedy action with respect to these sampled values:

$$a := argmax_i q(a_i)$$

Example

In the previous case, we had an estimate of how certain we were about the win-rate

Sampling from such distribution, allows us to give chances to the one that we only sampled 10 times to improve, similar to what UCB would so, but in a more grounded way.